EXPERIMENT-10

Implement Expectation Maximization algorithm using Gaussian Mixture Model.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.mixture import GaussianMixture

print("="*70)
print("      EXPECTATION MAXIMIZATION")
print("="*70)

uploaded=files.upload()

data=pd.read_csv("Iris.csv")

print("\nDataset")

print(data.head(10))

X=data.iloc[:,1:5]

model=GaussianMixture(
n_components=3,
random_state=42)

print("\nTraining EM Model...")
```

```
model.fit(X)

clusters=model.predict(X)

print("\nCluster Labels")

print(clusters)
print("\nFirst 20 Cluster Assignments")

for i in range(20):
    print("Record",i+1,"-> Cluster",clusters[i])

print("\nCluster Means")
print(model.means_)
print("\nCluster Weights")
print(model.weights_)

sample=[[5.1,3.5,1.4,0.2]]
result=model.predict(sample)

print("\nNew Sample")
print(sample)
print("\nAssigned Cluster")
print(result[0])
```

Sample Input:

```
...  ======================================================================
           EXPECTATION MAXIMIZATION
     ======================================================================
     Choose Files  Iris.csv
     Iris.csv(text/csv) - 5107 bytes, last modified: 8/5/2026 - 100% done
     Saving Iris.csv to Iris (3).csv
```

Sample Output:

```
Dataset
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0   1            5.1           3.5            1.4           0.2  Iris-setosa
1   2            4.9           3.0            1.4           0.2  Iris-setosa
2   3            4.7           3.2            1.3           0.2  Iris-setosa
3   4            4.6           3.1            1.5           0.2  Iris-setosa
4   5            5.0           3.6            1.4           0.2  Iris-setosa
5   6            5.4           3.9            1.7           0.4  Iris-setosa
6   7            4.6           3.4            1.4           0.3  Iris-setosa
7   8            5.0           3.4            1.5           0.2  Iris-setosa
8   9            4.4           2.9            1.4           0.2  Iris-setosa
9  10            4.9           3.1            1.5           0.1  Iris-setosa

Training EM Model...

Cluster Labels
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 0 2 0 2 0 2
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 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0]

First 20 Cluster Assignments
Record 1 -> Cluster 1
Record 2 -> Cluster 1
Record 3 -> Cluster 1
Record 4 -> Cluster 1
Record 5 -> Cluster 1
Record 6 -> Cluster 1
Record 7 -> Cluster 1
Record 8 -> Cluster 1
Record 9 -> Cluster 1
Record 10 -> Cluster 1
Record 11 -> Cluster 1
Record 12 -> Cluster 1
Record 13 -> Cluster 1
Record 14 -> Cluster 1
Record 15 -> Cluster 1
Record 16 -> Cluster 1

Record 17 -> Cluster 1
Record 18 -> Cluster 1
Record 19 -> Cluster 1
Record 20 -> Cluster 1

Cluster Means
[[6.54639415 2.94946365 5.48364578 1.98726565]
 [5.006      3.418      1.464      0.244     ]
 [5.9170732  2.77804839 4.20540364 1.29848217]]

Cluster Weights
[0.36539574 0.33333333 0.30127092]

New Sample
[[5.1, 3.5, 1.4, 0.2]]

Assigned Cluster
1
```